

Brief Phase 1 Validation Report: ETF Walk-Forward Results

Objective

The goal of Phase 1 was to validate the current ETF Kalman baseline across `SPY`, `QQQ`, and `SPYG` using year-by-year walk-forward results. The model was tested with fixed parameters:

```
q_jerk = 1e-7;
r_meas = 5e-4;

buySignal = trend_est > 0;
sellSignal = trend_high < 0;

buyConfirmDays = 5;
sellConfirmDays = 8;
transactionCost = 0.001;
```

The purpose was not to retune the model, but to check whether the strategy consistently improves return, drawdown, and Sharpe ratio across different ETF market regimes.

Main Findings

The model's strongest value appears during market stress years, especially `2020` and `2022`.

Across all three ETFs in `2020`, the strategy strongly outperformed buy-and-hold. Average excess return was about `+20.05 percentage points`, average drawdown improvement was about `+17.91 percentage points`, and average Sharpe improvement was about `+0.96`.

In `2022`, the strategy again helped across all three ETFs. Average excess return was about `+7.20 percentage points`, average drawdown improvement was about `+4.77 percentage points`, and average Sharpe improvement was about `+0.10`.

This suggests the Kalman model is most useful as a regime-risk filter rather than as a constant return enhancer.

Year-by-Year Pattern

In smooth bull-market years, the strategy often matched or slightly lagged buy-and-hold. This makes sense because, during strong upward trends, the best possible behavior is usually to stay fully invested. Any defensive exit can cause the model to miss part of the upside.

In crash or bear-market years, the strategy performed much better. The model reduced exposure during negative regimes and helped limit drawdowns.

The key pattern is:

Market Type	Model Behavior
Strong bull market	Usually matches or slightly trails buy-and-hold
Crash/recovery year	Often improves return and drawdown
Bear-market year	Often reduces losses
Sideways/noisy year	Can lag due to unnecessary exits

ETF-Specific Observations

SPY appears to be the cleanest fit for the current model. It had fewer trades and mostly behaved like buy-and-hold during normal years, while improving performance during major stress periods.

SPYG also fits the current model well. It had strong improvements in **2020** and **2022**, and otherwise often stayed close to buy-and-hold.

QQQ is more sensitive. It benefits during drawdown years, but it more often lags during strong growth years. This supports the earlier observation that QQQ may prefer slightly different tuning or a smoother exposure system.

Aggregate Validation Summary

Across all ETF-year results from **2018** through **2026**:

Metric	Result
Return wins	6
Return losses	9
Return ties	12
Drawdown wins	7
Drawdown losses	3
Drawdown ties	17
Sharpe wins	7
Sharpe losses	8
Sharpe ties	12

Metric	Result
Average excess return	+1.70 percentage points
Average drawdown improvement	+2.33 percentage points
Average Sharpe improvement	+0.077

These results show that the strategy does not beat buy-and-hold every year. However, it does appear to add value during the years where risk control matters most.

Conclusion

The current Kalman ETF model should be treated as a drawdown and regime-risk management model, not simply a return-maximizing model.

The strategy's main advantage is that it can reduce exposure during major negative regimes while staying mostly invested during normal upward markets. The model is worth continuing, but the next improvements should focus on making exits less binary and reducing the cost of being slightly wrong.

The best next development direction is partial exposure rather than more parameter tuning.